

Day 42 — TNPSC Group 2A Study Material

Part A – General Studies: Land Reforms, Agriculture, and Application of Science & Technology in Agriculture

READING MATERIAL

Covers land reform's core objective, the Green Revolution and M.S. Swaminathan's role, several Tamil-origin scientists tested via match-the-following, and a cluster of agriculture-focused government schemes.

Q: Real exam: What is the real objective of Tenancy reform (a category of Land Reform)?

A: Abolition of Intermediaries — confirmed by a real exam question, distinguishing it from other plausible-sounding but incorrect options like 'increase rural literacy' or 'eradication of rural indebtedness' (those relate to other rural development goals, not tenancy reform specifically).

Q: Real exam: Which statements about M.S. Swaminathan are TRUE — Father of Green Revolution in India; Agronomist and Plant Geneticist; wrote the book 'Cotton'; pioneer in mutation plant breeding?

A: THREE of the four are true: he is regarded as the Father of the Green Revolution in India; he was an agronomist and plant geneticist; and he was a pioneer in mutation plant breeding. He did NOT write a book called 'Cotton' — that claim is false, confirmed by a real exam question testing exactly this set of statements.

Q: Real exam: Match these Tamil-origin scientists to their fields — Sir C.V. Raman, Venki Ramakrishnan, Myswamy Annadurai, M.S. Swaminathan.

A: Sir C.V. Raman (Chandrasekhara Venkata Raman) → Light scattering (discovered the Raman Effect). Venki Ramakrishnan → Function of ribosomes (Nobel Prize in Chemistry, 2009). Myswamy Annadurai → Chandrayaan-1 (ISRO's first lunar mission; he was Project Director). M.S. Swaminathan → Green Revolution in India — all four confirmed correct via a real exam's match-the-following question.

Q: Real exam: Has Tamil Nadu declared the Cauvery Delta Zone a protected agricultural zone, and is this directly explained by a specific budget allocation?

A: Yes — Tamil Nadu declared the Cauvery Delta Zone a Protected Special Agricultural Zone (TRUE). Separately, the Agriculture Budget 2023-24 allotted ₹1,000 Crores to build infrastructure in the Cauvery Delta Zone (also TRUE) — but a real exam confirmed this budget allocation is NOT the correct EXPLANATION for why the zone was declared protected; the two facts are both true but not causally linked the way a simple Assertion-Reason pairing might suggest.

Q: Real exam: Match these agriculture-focused government schemes to their tested focus area — National Food Security Mission, Rainfed Area Development Programme, National Agricultural Development Programme (RKVY), Drought Prone Area Programme.

A: National Food Security Mission → increasing production of rice and pulses. Rainfed Area Development Programme → improving rainfed farming. National Agricultural Development Programme (RKVY) → promotion of horticulture in rainfed areas. Drought Prone Area Programme → augmenting mechanisation in agriculture — all four confirmed via a real exam's match-the-following question (note: these pairings reflect the exam's specific tested framing, which may emphasise one aspect of a broader scheme rather than its full textbook description).

MOCK TEST (WITH ANSWERS)

1. The objective of Tenancy reform (a category of Land Reform) is:

- A. Abolition of Intermediaries ✓**
- B. Increase Rural Literacy
- C. Regulation of Money Lenders
- D. Eradication of Rural Indebtedness

Explanation: Abolition of Intermediaries. [medium, question-bank-2022]

2. Which of the following statements about M.S. Swaminathan is FALSE?

- A. He was the Father of the Green Revolution in India
- B. He was an Agronomist and Plant Geneticist
- C. He wrote the book 'Cotton' ✓**
- D. He was a pioneer in mutation plant breeding

Explanation: He did not write a book called 'Cotton' — this claim is false. [hard, question-bank-2026]

3. Sir C.V. Raman is famous for his work on:

- A. Function of ribosomes
- B. Light scattering ✓**
- C. Chandrayaan-1
- D. Green Revolution in India

Explanation: Light scattering (Raman Effect). [easy, question-bank-2026]

4. Venki Ramakrishnan won the Nobel Prize for his work on:

- A. The Raman Effect
- B. Function of ribosomes ✓**
- C. Chandrayaan-1
- D. Green Revolution in India

Explanation: Function of ribosomes. [medium, question-bank-2026]

5. Myswamy Annadurai is best known for being Project Director of:

- A. The Green Revolution
- B. Chandrayaan-1 ✓**
- C. The discovery of the Raman Effect
- D. The Mandal Commission

Explanation: Chandrayaan-1. [medium, question-bank-2026]

6. Tamil Nadu has declared the Cauvery Delta Zone as a:

- A. Special Economic Zone
- B. Protected Special Agricultural Zone ✓**
- C. Industrial Corridor
- D. Coastal Regulation Zone

Explanation: Protected Special Agricultural Zone. [medium, question-bank-2026]

7. In the Agriculture Budget 2023-24, how much did the Tamil Nadu Government allot to build infrastructure in the Cauvery Delta Zone?

- A. ₹500 Crores
- B. ₹1,000 Crores ✓**
- C. ₹2,000 Crores
- D. ₹5,000 Crores

Explanation: ₹1,000 Crores. [hard, question-bank-2026]

8. The National Food Security Mission's tested focus, per a real exam, is:

- A. Augmenting mechanisation in agriculture
- B. Improving rainfed farming
- C. Increasing production of rice and pulses ✓**
- D. Promotion of horticulture in rainfed areas

Explanation: Increasing production of rice and pulses. [hard, question-bank-2026]

9. The Rainfed Area Development Programme's tested focus is:

- A. Augmenting mechanisation in agriculture
- B. Improving rainfed farming ✓**
- C. Increasing production of rice and pulses
- D. Promotion of horticulture in rainfed areas

Explanation: Improving rainfed farming. [medium, question-bank-2026]

Part B – Aptitude & Mental Ability: Time and Work

READING MATERIAL

Time and Work is the last of the 10 named Aptitude skills — after this, Part B moves to its Reasoning unit. Real exam evidence shows heavy use of 'trap' questions designed to look like they need complex math but actually resolve via a simple ratio.

Q: Method — inverse proportion (more workers, less time): 12 men complete a work in 36 days. In how many days will 18 men finish the same work?

A: Men $\times$ Days = constant (total work). $12 \times 36 = 18 \times x \rightarrow x = 432/18 = 24$ days. More workers means proportionally fewer days.

Q: Method — sequential combined work: A, B, C can complete a job alone in 8, 12, 16 days respectively. A and C work together for 2 days, then C leaves and B joins A. In how many more days do A and B finish?

A: Work rates: $A=1/8$, $B=1/12$, $C=1/16$ per day. In 2 days, A+C complete $2 \times (1/8 + 1/16) = 2 \times (3/16) = 3/8$ of the work. Remaining = $5/8$.
A+B's combined rate = $1/8 + 1/12 = 5/24$ per day. Days needed = $(5/8) \div (5/24) = (5/8) \times (24/5) = 3$ days.

Q: Method — wages split by share of work done: A can do a job in 10 days, B in 15 days. They work together for 5 days, then C finishes the remaining work in 2 days. Total payment for the whole job is Rs.5,000 — how should it be split?

A: Pay is split in proportion to how much of the TOTAL work each person actually did, not how many days they worked. In 5 days, A completes $5/10=1/2$ of the job, B completes $5/15=1/3$; together that's $5/6$, leaving $1/6$ for C to finish. Split of Rs.5,000 by work share: A gets $1/2 \times 5000 = \text{Rs.}2,500$; B gets $1/3 \times 5000 \approx \text{Rs.}1,667$; C gets $1/6 \times 5000 \approx \text{Rs.}833$. The key insight is the split follows WORK DONE, not days worked — C worked fewer days than B but did a comparable or different share depending on the rates, and pay tracks output, not time.

Q: Method — pairwise combined rates $\rightarrow$ solve for all three together: A+B finish a job in 12 days, B+C in 15 days, C+A in 20 days. In how many days can A, B, C finish it together?

A: Add all three pairwise rates: $(A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 2(A+B+C)$'s rate. LCM=60: $5/60 + 4/60 + 3/60 = 12/60 = 1/5$.
So $2 \times (A+B+C \text{ rate}) = 1/5 \rightarrow (A+B+C) \text{ rate} = 1/10 \rightarrow$ together they finish in 10 days.

Q: Trap question: If 9 spiders make 9 webs in 9 days, how many days does 1 spider take to make 1 web?

A: 9 days — NOT 1 day. Each spider makes 1 web in 9 days regardless of how many spiders work in parallel (they're not collaborating on the SAME web). This tests whether you recognize the rate-per-individual stays fixed, the classic trap being to assume the answer scales down to 1.

Q: Method — pipes filling and draining together: Two pipes fill a tank in 30 and 40 minutes separately; a third pipe drains it in 24 minutes. With all three open, how long to fill the tank?

A: Combined rate = $1/30 + 1/40 - 1/24$ (subtract the draining pipe's rate). LCM=120: $4/120 + 3/120 - 5/120 = 2/120 = 1/60$. So the tank fills in 60 minutes.

Q: Method — provisions lasting fewer days with more people (inverse proportion again): 1,000 soldiers have 70 days of provisions. If 400 more soldiers join, how long will provisions last?

A: People $\times$ Days = constant (total provisions). $1000 \times 70 = 1400 \times x \rightarrow x = 70000/1400 = 50$ days.

Q: Method — find the second worker's solo time from a combined time and the first worker's solo time: A and B together finish a job in 16 days; A alone takes 48 days. How long does B alone take?

A: B's rate = $(A+B)$'s rate - A's rate = $1/16 - 1/48 = 3/48 - 1/48 = 2/48 = 1/24$. So B alone takes 24 days.

Q: Trap question: If 5 persons complete 5 projects in 5 days, how long do 50 persons take to complete 50 projects?

A: Still 5 days — NOT 50 days. The ratio of people to projects stays the same (1:1), so the per-project time doesn't change; 50 people working on 50 projects is just 10 independent copies of the original 5-person/5-project scenario happening in parallel.

MOCK TEST (WITH ANSWERS)

1. 15 men complete a work in 24 days. In how many days will 20 men finish the same work?

- A. 16 days
- B. 18 days ✓**
- C. 20 days
- D. 30 days

Explanation: $15 \times 24 = 20 \times x \rightarrow x = 360/20 = 18$ days. [easy, authored]

2. A and B together can finish a job in 18 days; A alone takes 45 days. How long does B alone take?

- A. 24 days

- B. 27 days
- C. 30 days ✓**
- D. 36 days

Explanation: B's rate = $\frac{1}{18} - \frac{1}{45} = \frac{5}{90} - \frac{2}{90} = \frac{3}{90} = \frac{1}{30}$. B alone takes 30 days. [medium, authored]

3. If 7 workers build 7 walls in 7 days, how many days does 1 worker take to build 1 wall?

- A. 1 day
- B. 7 days ✓**
- C. 14 days
- D. 49 days

Explanation: 7 days — each worker's individual rate is unaffected by how many work in parallel on separate walls. [medium, authored]

4. Two pipes fill a tank in 12 and 15 minutes separately; a third pipe drains it in 20 minutes. With all three open, how long to fill the tank?

- A. 8 minutes
- B. 10 minutes ✓**
- C. 12 minutes
- D. 15 minutes

Explanation: Rate = $\frac{1}{12} + \frac{1}{15} - \frac{1}{20}$. LCM=60: $\frac{5}{60} + \frac{4}{60} - \frac{3}{60} = \frac{6}{60} = \frac{1}{10}$. Time = 10 minutes. [medium, authored]

5. 800 soldiers have provisions for 60 days. If 200 more soldiers join, how long will the provisions last?

- A. 40 days
- B. 45 days
- C. 48 days ✓**
- D. 50 days

Explanation: $800 \times 60 = 1000 \times x \rightarrow x = 48000 / 1000 = 48$ days. [easy, authored]

6. A, B, and C can do a job alone in 24, 30, and 40 days respectively. Working together, how long will they take?

- A. 8 days
- B. 10 days ✓**
- C. 12 days
- D. 15 days

Explanation: Combined rate = $\frac{1}{24} + \frac{1}{30} + \frac{1}{40}$. LCM=120: $\frac{5}{120} + \frac{4}{120} + \frac{3}{120} = \frac{12}{120} = \frac{1}{10}$. Together they take 10 days. [medium, authored]

7. If 8 people complete 8 tasks in 8 days, how long do 40 people take to complete 40 tasks?

- A. 8 days ✓**
- B. 20 days
- C. 40 days
- D. 320 days

Explanation: Still 8 days — the people-to-task ratio (1:1) is unchanged, so per-task time is unchanged. [medium, authored]

Part C – General English: Prose: "The Nose Jewel" — C. Rajagopalachari

READING MATERIAL

A moral fable about Ramayya, a poor but honest man, told partly through the eyes of two sparrows nesting in his roof — one compassionate, one indifferent — who find a lost diamond nose-stud, setting up a test of Ramayya's honesty when his wife discovers it.

Q: Who wrote "The Nose Jewel", and who is Ramayya?

A: C. Rajagopalachari. Ramayya is a simple, poor-but-respectable man living in a modest tiled-roof house with his wife.

Q: What do the two sparrows nesting in Ramayya's roof represent, and how do their attitudes toward helping him differ?

A: The male sparrow is compassionate, asking 'Should we not help this poor Ramayya?' The female sparrow is dismissive/indifferent, telling him not to concern himself with matters that don't involve them, and to focus only on protecting their nest from the cat.

Q: What did the male sparrow find, where, and what did he do with it?

A: A diamond nose-stud, found in a muck-heap. He picked it up in his beak, brought it to Ramayya's house, and dropped it on the floor.

Q: Real exam: When/how did Ramayya's wife notice the jewel, and how did she feel?

A: While sweeping the floor — confirmed by a real exam question. She felt DELIGHTED upon finding it.

Q: What did Ramayya suggest doing with the nose-stud, and whose was it?

A: He suggested RETURNING it to its rightful owner rather than keeping it — reflecting his honesty despite his poverty. The stud belonged to Meenakshi Ammal's daughter, who had lost it.

Q: How did the female sparrow react to the jewel once it was brought to the nest/house?

A: With indifference — she remained unimpressed/uninterested, consistent with her earlier dismissive attitude toward helping Ramayya at all.

MOCK TEST (WITH ANSWERS)

1. "The Nose Jewel" was written by:

A. Munshi Premchand

B. C. Rajagopalachari ✓

C. R.K. Narayan

D. Ruskin Bond

Explanation: C. Rajagopalachari. [easy, web-research]

2. How did the female sparrow react to the diamond nose-stud?

A. She was excited

B. She was indifferent ✓

C. She was angry

D. She was curious

Explanation: She was indifferent. [medium, web-research]

3. Where did the male sparrow drop the diamond stud?

A. In the nest

B. In a bowl

C. On the floor ✓

D. On a shelf

Explanation: On the floor. [medium, web-research]

4. Who found the diamond stud in Ramayya's house?

A. Ramayya himself

B. Ramayya's wife ✓

C. The maid-servant

D. The village magistrate

Explanation: Ramayya's wife. [medium, web-research]

5. Real exam: When did Ramayya's wife notice the jewel?

A. While sweeping the floor ✓

B. While washing the floor

C. While dusting the floor

D. While mopping the floor

Explanation: While sweeping the floor. [hard, question-bank-2022]

6. How did Ramayya's wife feel when she found the diamond stud?

A. Delighted ✓

B. Disappointed

C. Anxious

D. Indifferent

Explanation: Delighted. [easy, web-research]

7. What did Ramayya suggest doing with the nose-stud?

A. Keeping it

B. Returning it to the owner ✓

C. Hiding it

D. Selling it

Explanation: Returning it to the owner. [medium, web-research]

8. Who had originally lost the nose-stud?

A. Meenakshi Ammal's daughter ✓

B. The village headman's wife

C. The maid-servant

D. The shopkeeper

Explanation: Meenakshi Ammal's daughter. [hard, web-research]